Section B Comprehensive Review - Mixed

Directions: Show all work. Use exact values. Box or circle all answers.

1. The table shows the number of milligrams of medication in a patient's system at the end of each hour during surgery.

Hour	1	2	3	4
Milligrams	2500	500	100	20

Find the common ratio and write a recursive formula to model the situation. Do not forget to include the first term.

2. A medication contains 4000 mg at the end of hour 1 and decreases by 30% each hour. Write a recursive formula.

3. Write the first six terms of a sequence if $a_1 = 2$ and $a_n = 5a_{n-1}$.

4. The table shows the number of milligrams of medication in a patient's system at the end of each hour during surgery.

Hour	1	2	3	4
Milligrams	3456	1152	384	128

Find the common ratio and write a recursive formula to model the situation. Do not forget to include the first term.

5. Write a recursive formula to represent the sequence $5, -15, 45, -135, \ldots$.

6. Write the first six terms of a sequence if $a_1 = 6$ and $a_n = 2a_{n-1}$.

7. Write a recursive formula to represent the sequence $96, -48, 24, -12, \ldots$.

8. A medication administered to a patient contains 1800 mg at the end of hour 1 and decreases by 10% each hour. Write a recursive formula to represent the amount of medication in the patient's system at any given time.

9. Write a recursive formula to represent the sequence $2, -8, 32, -128, \ldots$.

10. Write the first five terms of a sequence if $a_1 = 3$ and $a_n = -4a_{n-1}$.

11. Write a recursive formula to represent the sequence $7, 28, 112, 448, \ldots$.

12. Write the first five terms of a sequence if $a_1 = 6$ and $a_n = -2a_{n-1}$.

13. The table shows the mass of a sample at the end of each day.

Day	1	2	3	4
Mass	1250	1000	800	640

Find the common ratio and write a recursive formula to model the situation. Do not forget to include the first term.

14. Write the first five terms of a sequence if $a_1 = 243$ and $a_n = \frac{1}{3}a_{n-1}$.

15. Write a recursive formula to represent the sequence $28,14,7,\frac{7}{2}, \ldots$.

16. Write the first five terms of a sequence if $a_1 = 320$ and $a_n = \frac{1}{2}a_{n-1}$.

17. A population has 120 organisms at the end of hour 1 and increases by 20% each hour. Write a recursive formula.

18. Write a recursive formula to represent the sequence $20,10,5,\frac{5}{2}$, ...

19. Write the first six terms of a sequence if $a_1 = 81$ and $a_n = -\frac{1}{3}a_{n-1}$.

20. The table shows the mass of a sample at the end of each day.

Day	1	2	3	4
Mass	1296	864	576	384

Find the common ratio and write a recursive formula to model the situation. Do not forget to include the first term.

21. Write a recursive formula to represent the sequence $4,12,36,108$, ...

22. A population has 200 organisms at the end of hour 1 and increases by 15% each hour. Write a recursive formula.

23. Write a recursive formula to represent the sequence $81, -27, 9, -3, \ldots$.

24. Write the first six terms of a sequence if $a_1 = 128$ and $a_n = -\frac{1}{2}a_{n-1}$.